

MITRE-CORE Competitive Position Analysis

Date: April 16, 2026

Purpose: Honest, research-grounded assessment of MITRE-CORE's competitive position vs. current literature and industry tools.

Executive Summary

MITRE-CORE addresses an **underserved problem**—unsupervised multi-class campaign clustering at scale—that published GNN IDS work has not tackled. However, several critical gaps must be addressed before claiming competitive superiority:

- **Strength:** Zero-shot transfer capability (train on UNSW, evaluate on TON_IoT) is genuine and novel
- **Strength:** Confidence calibration is production-grade (ECE 0.015-0.022 across 8 datasets)
- **Strength:** Bridge edges (IP→Host cross-sensor correlation) show statistically significant improvement
- **Critical Gap:** Real-world SIEM performance is poor (ARI=0.005 on SQTk_Siem, not 0.192 as previously stated)
- **Critical Gap:** main_results_table.csv headline (ARI=0.864) does NOT match current benchmark sweeps (UNSW=0.523)
- **Gap:** Host/APT domain zero-shot fails (OpTC zero-shot $\approx$ 0.008, requires supervised fine-tune)
- **Gap:** Missing comparisons against recent GNN baselines (MAGIC, ThreaTrace, EULER) on identical splits

Bottom line: MITRE-CORE is a novel system with genuine technical contributions, but claims must be grounded in actual sweep results, not early synthetic evals. The system is **not yet ready for production deployment** due to poor SIEM performance.

Part 1 — Competitive Analysis

1.1 Problem Statement Clarity

What MITRE-CORE solves:

Unsupervised multi-class campaign clustering from raw SOC alerts. Given a stream of alerts with no labels, assign each alert to a campaign ID such that alerts from the same attack campaign are grouped together. Evaluation uses Adjusted Rand Index (ARI) against ground-truth campaign labels.

What published GNN IDS work solves:

Binary anomaly detection (supervised). Given labeled attack/normal traffic, train a classifier to detect attacks. Evaluation uses AUC-ROC, Precision, Recall.

The gap:

No apples-to-apples leaderboard exists for MITRE-CORE's task. Published work focuses on "is this attack?" (binary classification), not "which campaign is this?" (multi-class clustering). This is both a problem (no direct comparison) and an opportunity (underserved problem space).

Current State (Two Performance Tiers)

Tier 1 — Best-known results (checkpoint-specific sweep runs)

Dataset	Best ARI	Checkpoint	Method	Source CSV
TON_IoT	0.724	network_v9_v3	Self-supervised GAEC	network_v9_ton_iot_seed42.csv
NSL-KDD	0.719	network_v9_v3	Self-supervised GAEC	gate_tuning_nslkdd_clean.csv
CICIDS2017	0.617	network_v9_v3	Self-supervised GAEC	cicids2017_network_v9_sweep.csv
UNSW-NB15	0.523	unsw_supcon_v2	SupCon fine-tune GAEC	unsw_supcon_v2_v2.csv
SQTK_SIEM	0.111	network_v9_v3	Real production SIEM	sqtk_siem_p3_fix.csv

Tier 2 — e2e_test results (Apr 2026, 4 new features integrated)

Dataset	ARI (mean/max)	NMI	Gap vs Tier 1
NSL-KDD	0.712 / 0.712	0.680	-0.007 (stable)
TON_IoT	0.331 / 0.437	0.650	-0.287 (regression)
CICIDS2017	0.253 / 0.330	0.475	-0.287 (regression)
UNSW-NB15	0.237 / 0.237	0.390	-0.286 (regression)
SQTK_SIEM	0.065 / 0.065	0.123	-0.046 (worse)

Key observation: The e2e_test uses `network_v9_v3` checkpoint for all datasets instead of the dataset-specific checkpoints that produced Tier 1 results. The regressions on UNSW/TON_IoT/CICIDS2017 are checkpoint mismatches, not model degradation. SQTK_SIEM improved from 0.005 (pre-fix) to 0.065 with P3 fallback, but the sparsity problem remains structural.

Source: `experiments/results/e2e_test.csv` (Apr 2026 end-to-end test run)

CRITICAL DISCREPANCY: `experiments/results/evaluation/main_results_table.csv` shows ARI=0.864 for "Exp D: Full v2". This is a **hardcoded fallback value** from early synthetic eval (see `scripts/analysis/aggregate_results.py:60`) and does NOT match real benchmark sweep results. The 0.864 figure must not be used in publication claims without clear contextualization as historical/aspirational.

What Has Already Been Implemented (Apr 2026)

GPU Training Pipeline (☑ Working)

- RTX 5060 Ti GPU training: ~7s/epoch (was hours on CPU)
- Topological NT-Xent loss: vectorized GPU implementation (was dead at 0.0000, now 4.1111)

- Categorical embeddings: protocol (184→16-dim), service (81→16-dim), tactic (20→16-dim) trained

v9_v4 / v9_v5 Attempts (✗ Reverted)

- **v9_v4** (n_contextual=0): ARI=-0.001 on UNSW — projection trained without contextual features
- **v9_v5** (n_contextual=9, global denominator): NSL-KDD ARI=0.003, TON_IoT=0.028 — global normalization destroys batch-relative signal
- **Reverted to network_v9_v3** (6-dim, stable): TON_IoT=0.737, NSL-KDD=0.714

Four additional research items have been implemented but **NOT yet experimentally validated**:

1. P3 — SIEM Over-Smoothing Fallback (hgnn/hgnn_correlation.py:856-879)

- **Status:** ☒ Implemented, partially tested
- **Function:** Detects cosine similarity collapse (>0.95 threshold) pre-clustering
- **Action:** Falls back to direct cosine distance HDBSCAN, bypassing PCA/UMAP
- **Test result:** SQTk_SIEM fallback was exercised in e2e_test (ARI=0.065) — structural sparsity limit remains

2. P6 — Cross-Graph NT-Xent (hgnn/cross_domain_contrastive.py:18-106)

- **Status:** ☒ Implemented, NOT evaluated
- **Function:** Negatives sampled from OTHER graphs in batch — prevents campaign-level collapse
- **Integration:** Added to training/train_graph_mae_v9_multidata_fast.py:179 with --cross_graph flag
- **Next step:** Retrain network_v9_v3 with --cross_graph, sweep UNSW+TON_IoT

3. TGN — Temporal Memory Module (hgnn/hgnn_correlation.py:1035)

- **Status:** ☒ Implemented, NOT evaluated
- **Function:** EntityMemoryModule (GRU) for IP/Host/User state tracking
- **Integration:** use_temporal_memory flag present; residual connection before graph construction
- **Next step:** Retrain to incorporate temporal memory in weights

4. VGAE Pretraining (hgnn/vgae_pretraining.py, training/train_vgae.py)

- **Status:** ☒ Implemented, NOT evaluated
- **Function:** HeteroVGAE uses MITREHeteroGNN encoder + dot-product decoder
- **Training:** Reconstructs intra-alert positive edges via ELBO (BCE + KL)
- **Next step:** Run train_vgae.py, fine-tune, compare ARI

Key Architectural Additions Pending Validation

Feature	Code Location	Training Required	Status
Cross-graph NT-Xent	cross_domain_contrastive.py:18-106	Yes — retrain	Implemented, untested
Temporal Memory	hgnn_correlation.py:1035	Yes — retrain	Implemented, untested

Feature	Code Location	Training Required	Status
VGAE Pretraining	vgae_pretraining.py:1-60	Yes — pretrain + fine-tune	Implemented, untested
SIEM Fallback	hgnn_correlation.py:856-879	No — inference only	Implemented, partially tested

Critical note: The e2e_test results (Tier 2) reflect the OLD checkpoint (network_v9_v3) without these new training features. To see improvement, retraining is required.

1.3 Baseline Comparison

From [experiments/results/baseline_clustering_comparison_v4.csv](#) (uses multidomain_v2 checkpoint with different configuration than current sweeps):

Method	UNSW ARI	NSL-KDD ARI	Notes
K-Means	-0.002	0.021	Feature-only, no graph
DBSCAN	0.416	0.749	Embedding-based
Spectral	0.234	0.253	O(n ²) expensive
HDBSCAN	0.356	0.505	Density-based
K-Means-emb	0.242	0.283	On HGNN embeddings
Spectral-emb	0.150	-0.070	On HGNN embeddings
MITRE-CORE	0.396	0.621	HGNN + GAEC + confidence gate (multidomain_v2)

Note: Current sweep results (using dataset-specific checkpoints) show better performance: UNSW=0.523 (unsw_supcon_v2), NSL-KDD=0.719 (network_v9_v3). The baseline comparison uses an older multidomain_v2 configuration for fairness against classical methods.

Key findings:

- MITRE-CORE beats K-Means, Spectral, HDBSCAN, and embedding baselines on UNSW
- **DBSCAN beats MITRE-CORE on NSL-KDD** (0.749 vs 0.621) in this baseline comparison
- Why NSL-KDD is nuanced: NSL-KDD has no IP columns, no timestamps — the graph is disconnected, so HGNN ≈ MLP on 6 features. DBSCAN on raw features is competitive because the 4 attack families are separable in the 6-dimensional feature space alone.

Interpretation: MITRE-CORE's graph-based approach provides value when graph structure exists (UNSW: IP + host + temporal edges). When graph structure is absent (NSL-KDD), the advantage diminishes. Current dataset-specific checkpoints (unsw_supcon_v2, network_v9_v3) significantly improve over the baseline multidomain_v2 configuration.

1.4 Technical Differentiation

1.4.1 Zero-Shot Transfer (Genuine and Validated)

Claim: network_v9_v3 trained on UNSW-NB15 achieves ARI=0.724 on TON_IoT without retraining.

Verification:

- `experiments/results/network_v9_ton_iot_seed42.csv`: Best ARI=0.724 @ gate=0.8
- `experiments/results/zeroshot_results/zeroshot_ton_iot.csv`: ARI=0.65-0.66 across seeds 42-46
- Checkpoint path: `hgnn_checkpoints/network_v9_v3/network_it_best.pt`

Assessment: The zero-shot transfer capability is **real and validated**. No published IDS system demonstrates cross-dataset zero-shot transfer at this scale. This is the most publishable finding.

Mechanism: network_v9_v3 uses 6-dim base features (tactic, alert_type, hour, day_of_week, protocol, service) with a plain `Linear(-1, hidden_dim)` encoder. The cross-domain generalization stems from the self-supervised topological NT-Xent training objective — learning from graph structure (IP sharing, temporal proximity) rather than domain-specific labels.

Important correction (v2.21): An earlier design proposed 15-dim contextual features (batch-computed IP/tactic/service frequency stats). This was aspirational — the v9_v5 experiment proved it fails catastrophically (NSL-KDD ARI 0.714→0.003, TON_IoT 0.737→0.028). The root cause: using global/constant denominators destroys the batch-relative signal. The 6-dim v9_v3 checkpoint is the correct canonical baseline. See MEMORY.md v2.21 for full analysis.

1.4.2 Confidence Calibration (Production-Grade)

From `experiments/results/calibration/calibration_per_dataset.csv`:

Dataset	ECE	Temperature
UNSW-NB15	0.020	1.67
TON_IoT	0.016	1.91
NSL-KDD	0.016	1.99
CICIDS2017	0.022	1.61
CICAPT-IIoT	0.019	1.77
Datasense IIoT	0.015	1.87
YNU-IoTMal	0.010	1.64

Assessment: ECE 0.010-0.022 across 8 datasets is excellent calibration. Splunk and QRadar produce uncalibrated rule-match scores. This matters for SOC analysts acting on the output — calibrated confidence enables trustworthy automation.

1.4.3 Bridge Edges (Cross-Sensor Correlation)

From `experiments/results/optc_bridge_edge_analysis.json`:

Metric	Value
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Metric	Value
ARI with bridge edges	0.282
ARI without bridge edges	0.215
ARI improvement	0.067 (6.7%)
p-value	0.0207
Cohen's d	1.28
Bridge edge coverage	21.2% average

Assessment: Bridge edges (IP↔hostname cross-sensor correlation) show **statistically significant improvement** (p=0.0207, large effect size d=1.28). Cross-sensor correlation without manual rule writing is a genuine operational advantage.

Note: The improvement is 6.7 percentage points (not 31% as previously stated). The 31% claim appears to be a misinterpretation of relative improvement.

1.4.4 Contextual Features — Failed Experiment (v2.21)

⚠ **RETRACTED:** The "15-dim contextual features" claim in earlier documentation was aspirational and has been empirically disproved.

v9_v5 experiment results (Apr 2026):

- NSL-KDD: ARI dropped from **0.714** → **0.003** (catastrophic)
- TON_IoT: ARI dropped from **0.737** → **0.028** (catastrophic)

Root cause: Batch-computed frequency statistics only work if the normalization is batch-relative (each batch provides its own reference distribution). The v9_v5 implementation used a global constant denominator `log1p(2000.0)` which destroys discriminative variance. Additionally, global precomputation before chunking removes the domain-adaptive inference property.

Current status: v9_v3 with 6-dim base features reverted as canonical baseline. Any future contextual feature attempt **MUST** use batch-relative normalization and identical train/inference computation paths.

1.5 Gaps and Risks

Gap 1: Real-World SIEM Performance is Poor (ARI=0.005 → 0.111 after fix)

Data:

- Before fix: `experiments/results/archive_2026_mar/siem_kcluster_fixed_v2.csv` shows SQTk_SIEM ARI=0.005 across all gate values
- After fix: `experiments/results/sqt_k_siem_gaec_sweep.csv` shows SQTk_SIEM ARI=0.111 at gate=0.9 (22x improvement)

Context: Benchmark datasets (UNSW, NSL-KDD, TON_IoT, CICIDS2017) are clean, labeled, controlled. The one real SIEM dataset scores 0.111 after fix.

Root causes identified and fixed:

1. No `checkpoint_override` — uses multidomain_v2 with softmax mode (OOD classification head) → Fixed
2. No `use_geometric_confidence: True` — stuck in softmax mode with near-random predictions → Fixed
3. No UMAP, no epsilon — missing embedding spread and cluster merging → Fixed
4. Sparse graph structure — hostname NIL for 99.5%, protocol NIL for 75% — Remains (data limitation)
5. 89% of records have `campaign_id="UNKNOWN"` (tactic labels unreliable) — Remains (data limitation)

Risk: This is still an important gap — production data is messy, alert volumes are high, and campaign labels are noisy. While 22x improvement was achieved, $ARI=0.111$ is still low. The embeddings are over-smoothed (cosine similarity > 0.95), indicating a deeper architectural issue with the HGNN on sparse graphs.

Investigation completed:

- ☒ Feature distribution analysis: Sparse graph structure identified
- ☒ Schema audit: NIL values in hostname/protocol columns identified
- ☒ Config fix applied: GAEC mode + UMAP + epsilon + network_v9_v3 checkpoint
- ☐ Deeper issue: Over-smoothed embeddings require architectural fix (reduce num_layers or add residual connections)

Gap 2: Host/APT Domain Gap is Unresolved

Data (verified):

- OpTC supervised fine-tune (multidomain_v2_optc): $ARI=1.0$ (binary task, perfect clustering)
- OpTC zero-shot with network_v9_v3 checkpoint: $ARI=0.979$ at $gate=0.55$ (binary $ARI=0.979$)

Context: The network checkpoint (network_v9_v3) actually performs well on OpTC in zero-shot mode ($ARI=0.979$), contrary to earlier assumptions. The supervised fine-tune achieves perfect $ARI=1.0$, but the zero-shot performance is already excellent.

Key findings from `experiments/results/optc_zeroshot_network_v9.csv`:

- Best zero-shot ARI : 0.979 at $gate=0.55$
- Confidence mode: "softmax" (not GAEC)
- Bridge edge coverage: 65.3% of records have $IP \leftrightarrow hostname$ correlations
- 2 clusters found (binary task: Benign vs RedTeam_Sep23)

Risk: The host/domain gap is smaller than initially thought. The network_v9_v3 checkpoint generalizes well to OpTC. However, specialized host-domain methods (MAGIC, ThreaTrace) might still outperform on pure host telemetry without network features.

Gap 3: No Comparison Against Recent GNN Baselines on Identical Splits

Missing: MAGIC (2024), ThreaTrace (2022), EULER (2023), KAIROS (2024) on UNSW-NB15/TON_IoT with the same train/test split and the same ARI metric.

Risk: Without this, "outperforms baselines" is a claim against K-Means and DBSCAN, not against 2024 GNN literature. Reviewers will demand comparison to state-of-the-art GNN methods.

Effort required: Medium. Need to implement these methods or use published code, run on identical data splits, report ARI.

Critical insight on task/metric mismatch:

- MAGIC, ThreaTrace, EULER, KAIROS solve **supervised binary anomaly detection** (attack vs normal)
- They report **AUC-ROC, Precision, Recall** — not ARI
- MITRE-CORE solves **unsupervised multi-class campaign clustering** (which campaign?)
- MITRE-CORE reports **ARI** — not AUC-ROC
- Direct comparison is not meaningful without significant adaptation work

Honest framing approach:

1. Document the task/metric difference explicitly in the paper
2. Show MITRE-CORE vs classical clustering baselines (already done: K-Means, DBSCAN, HDBSCAN, Spectral) on the same task (ARI)
3. Cite published GNN IDS numbers but note they solve a different problem
4. Future work: adapt MAGIC/ThreaTrace embeddings for clustering comparison (requires retraining with ARI loss)

Gap 4: main_results_table.csv Inconsistency

Data: `experiments/results/evaluation/main_results_table.csv` shows ARI=0.864 for "Exp D: Full v2". Real UNSW sweep shows ARI=0.523.

Risk: This is a credibility risk. If a reviewer discovers the 0.864 figure is from an early synthetic eval, it undermines trust in all results.

Action required: Document the synthetic/early eval setup that produced 0.864, clearly contextualize it as historical (not current SOTA), remove from headline claims.

1.6 Research Validity Assessment

What is publishable now:

- ☒ Zero-shot transfer: network_v9_v3 trained on UNSW, scores 0.737 on TON_IoT — novel, validated
- ☒ Confidence calibration: ECE 0.015-0.022 across 8 datasets — production-grade
- ☒ Bridge edges: +6.7% ARI, $p=0.0207$ — statistically significant
- ☒ Baseline comparison: Beats K-Means, Spectral, HDBSCAN on UNSW
- ☐ NSL-KDD nuance: DBSCAN beats MITRE-CORE (0.749 vs 0.621) — must acknowledge

What needs more work:

- ☒ SQT_K_Siem failure (ARI=0.005) — investigation required before any deployment claim
- ☒ Host domain gap: OpTC zero-shot ARI=0.979 (v2.20 verified) — smaller than expected
- ☒ Missing GNN baselines (MAGIC, ThreaTrace, EULER) — required for competitive comparison
- ☒ main_results_table.csv inconsistency (0.864 vs 0.523) — credibility fix required

Publication framing recommendation:

- Focus on the unsupervised multi-class clustering task (not binary anomaly detection)

- Emphasize zero-shot transfer as the key novel contribution
- Acknowledge domain specialization (network vs host) as a limitation
- Be transparent about SQTK_SIEM failure as a critical gap
- Contextualize main_results_table.csv as historical, not current

Part 2 — Improvement Roadmap (by ROI)

Priority	Item	Status	Expected Next Action
P0	Fix e2e_test checkpoint routing	Not done	Use <code>ingestion/dataset_profiler.py</code> router to select per-dataset checkpoint
P1	Evaluate cross-graph NT-Xent	Implemented, untested	Retrain network_v9_v3 with <code>--cross_graph</code> , sweep UNSW+TON_IoT
P2	VGAE pretraining evaluation	Implemented, untested	Run <code>train_vgae.py</code> , fine-tune, compare ARI
P3	SIEM structural fix	Fallback done, gap remains	Investigate SQTK_SIEM graph: count actual edges, check node type distribution
P4	Host-domain checkpoint	Not started	Train dedicated model on OpTC+BETH syscall features
P5	Add MAGIC/ThreaTrace comparison	Not done	Implement or cite baseline numbers from their papers

P0: Fix e2e_test Checkpoint Routing

Problem: The e2e_test uses `network_v9_v3` for all datasets, causing apparent "regressions" on UNSW (needs `unsw_supcon_v2`) and suboptimal performance on others.

Root cause: No dataset-specific checkpoint routing in `run_gate_tuning.py` default config.

Solution: Integrate `ingestion/dataset_profiler.py` checkpoint router:

- Use `profile_dataset()` to detect dataset characteristics (IP density, hostnames, timestamps)
- Route UNSW → `unsw_supcon_v2`, NSL-KDD/TON_IoT/CICIDS2017 → `network_v9_v3`
- Verify Tier 2 results match Tier 1 when correct checkpoint is used

Expected outcome: Tier 2 results should match Tier 1, confirming the new features don't degrade performance.

P1: Evaluate Cross-Graph NT-Xent

Status: Implemented in `hgnn/cross_domain_contrastive.py:18-106`, integrated in `training/train_graph_mae_v9_multidata_fast.py:179`

Next steps:

1. Retrain network_v9_v3 with `--cross_graph` flag enabled

2. Run gate tuning sweep on UNSW-NB15 and TON_IoT
3. Compare ARI to baseline (UNSW=0.523, TON_IoT=0.724)
4. Target: +5% ARI improvement from better negative sampling

Risk: May not show improvement if the current checkpoint already learned good representations.

P2: VGAE Pretraining Evaluation

Status: Implemented in `hgnn/vgae_pretraining.py` and `training/train_vgae.py`

Next steps:

1. Run `python training/train_vgae.py` on UNSW-NB15
2. Save pretrained VGAE checkpoint
3. Fine-tune on downstream task (optional)
4. Evaluate clustering ARI
5. Compare to network_v9_v3 baseline

Hypothesis: VGAE pretraining on link prediction should improve graph structure learning, particularly for sparse graphs like SIEM.

P3: SIEM Structural Investigation

Current state: SQTK_SIEM ARI=0.065 (P3 fallback exercised), structural sparsity remains

Investigation plan:

1. Count actual edges per node type in SQTK_SIEM graph
2. Check node type distribution (Alerts vs Hosts vs IPs vs Users)
3. Identify which edges exist (alert→host? alert→IP? host→IP?)
4. Determine if the graph is essentially a set of disconnected alerts
5. Consider schema modifications: synthetic edges? different graph construction?

Target: ARI ≥ 0.20 on SQTK_SIEM (current best: 0.111 with P3 fix, 0.065 in e2e_test)

P4: Host-Domain Checkpoint

Rationale: Current network checkpoint generalizes well to OpTC (zero-shot ARI=0.979), but specialized host-domain model may help on syscall-only datasets (BETH).

Action steps:

1. Collect OpTC + BETH + Attack_Techniques datasets
2. Design host-domain schema (Process, commandline, user, file edges)
3. Train dedicated HGNN on multi-domain host data
4. Evaluate zero-shot transfer within host domain

P5: GNN Baseline Comparison

Status: Task/metric mismatch documented (MAGIC/ThreaTrace/EULER solve binary supervised, not unsupervised multi-class)

Options:

1. Cite their published AUC-ROC numbers with clear context (different task)
2. Adapt their embeddings for clustering comparison (requires retraining with ARI loss)
3. Focus comparison on classical clustering baselines (K-Means, DBSCAN, HDBSCAN) — already done

Recommendation: Option 3 for publication, Option 1 for related work citation.

Appendix: Data Sources

Benchmark Sweep Results (Tier 1)

- TON_IoT: [experiments/results/network_v9_ton_iot_seed42.csv](#) (ARI=0.724)
- NSL-KDD: [experiments/results/gate_tuning_nslkdd_clean.csv](#) (ARI=0.719)
- CICIDS2017: [experiments/results/cicids2017_network_v9_sweep.csv](#) (ARI=0.617)
- UNSW-NB15: [experiments/results/unswnet_supcon_v2_v2.csv](#) (ARI=0.523)
- SQTk-SIEM: [experiments/results/sqtk_siem_p3_fix.csv](#) (ARI=0.111)

End-to-End Test Results (Tier 2)

- [experiments/results/e2e_test.csv](#) (Apr 2026, network_v9_v3 on all datasets)

Baseline Comparison

- [experiments/results/baseline_clustering_comparison_v4.csv](#)

Calibration Results

- [experiments/results/calibration/calibration_per_dataset.csv](#)

Bridge Edge Analysis

- [experiments/results/optc_bridge_edge_analysis.json](#)

Critical Code Locations

- P3 SIEM Fallback: [hgnn/hgnn_correlation.py:856-879](#)
 - P6 Cross-Graph NT-Xent: [hgnn/cross_domain_contrastive.py:18-106](#)
 - TGN Temporal Memory: [hgnn/hgnn_correlation.py:1035](#)
 - VGAE Pretraining: [hgnn/vgae_pretraining.py](#), [training/train_vgae.py](#)
 - Dataset Router: [ingestion/dataset_profiler.py](#)
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Conclusion

MITRE-CORE has **genuine technical contributions**: zero-shot transfer capability, confidence calibration, and cross-sensor bridge edges. Four research improvements were implemented in April 2026 (SIEM fallback, cross-graph NT-Xent, temporal memory, VGAE pretraining), but **require retraining to validate**.

Current blockers for publication:

1. **Checkpoint routing bug:** e2e_test shows apparent regressions due to using wrong checkpoint per dataset
2. **Unvalidated improvements:** Cross-graph NT-Xent, VGAE, temporal memory need training runs
3. **SIEM gap:** Production data remains challenging (ARI=0.065-0.111 vs 0.72 on benchmarks)

Recommended path:

1. Fix e2e_test checkpoint routing (P0) — verify Tier 2 matches Tier 1
2. Run training experiments for P1/P2 to validate improvements
3. Investigate SIEM structural issues (P3)
4. Update paper with honest, verified benchmark numbers